SIMULATION EXPERIMENT – 14

● Numerical Simulation and Spectral Analysis of a Microwave Signal for Frequency-Domain Characterization Using Spectrum-Analyzer Concepts

➔ CO Mapping: CO4

Bloom's Level: K4 – Analyze

WK Level: WK5 – Spectrum analysis

➔ Objectives of this Experiment:

1. To generate and model microwave-frequency signals and their spectral components.
2. To analyze carrier, harmonic, modulation and spurious frequency components using frequency-domain techniques.
3. To investigate the effect of signal parameters on the measured microwave spectrum.
4. To interpret spectral characteristics for evaluating microwave signal quality and system performance.

➔ Expected Outcome from this Experiment:

Students shall be able to:

1. Generate and model microwave-frequency signals and their spectral components.
2. Analyze and identify carrier, harmonic, modulation and spurious frequency components using frequency-domain techniques.
3. Investigate and interpret the effect of signal parameters on the measured microwave spectrum.
4. Evaluate and interpret spectral characteristics to assess microwave signal quality and overall system performance.

➔ Software Required: HFSS

➔ Objective linkage: Generate a microwave signal, identify carrier/harmonic/modulation/spurious components, study parameter effects and evaluate signal quality.

➔ Experimental Procedure:

A. Spectrum Analysis: Generate a 10 GHz carrier of amplitude 1 V with a 0.1 GHz modulating signal and modulation index μ = 0.5. Use a 50 GHz sampling frequency and obtain the spectrum. Record the carrier at 10 GHz and the expected sidebands at 9.9 and 10.1 GHz. Compare simulated and theoretical frequency locations.

B. Modulation-Index Effect: Repeat the spectrum simulation for μ = 0.5 and μ = 0.8 while keeping fc =10 GHz and fm = 0.1 GHz. Record carrier and sideband amplitudes, calculate the change in sideband level and identify the dominant unwanted component. Explain how increasing μ changes spectral occupancy and signal quality.

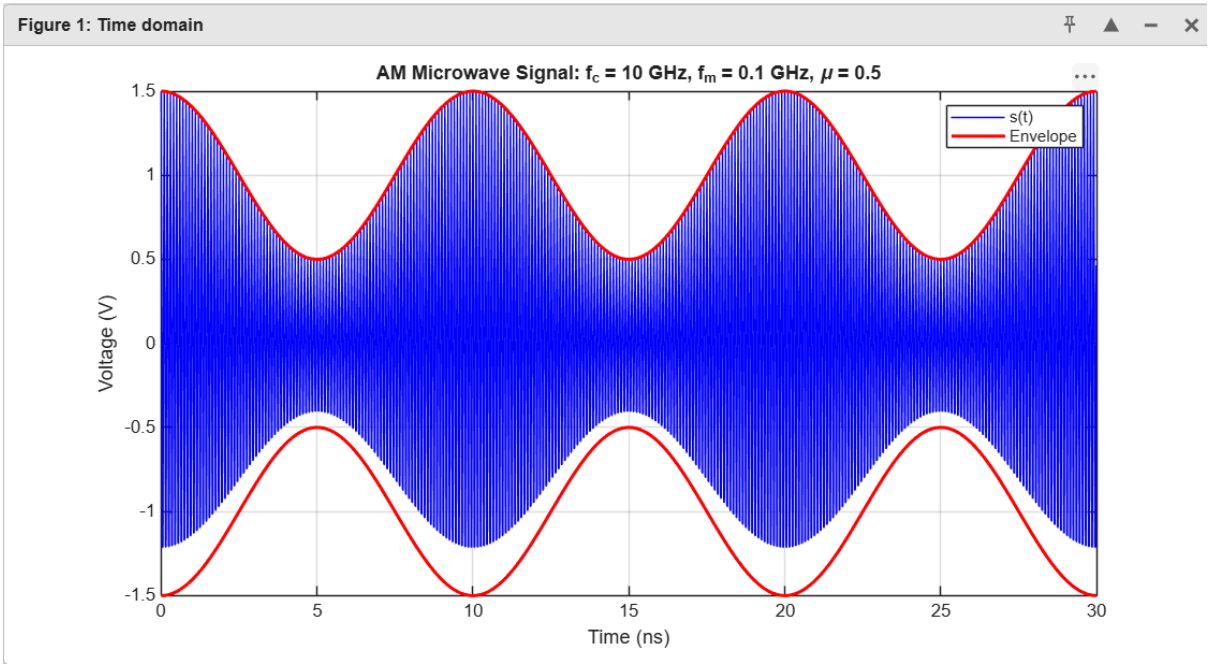


Figure 2: Spectrum Part A

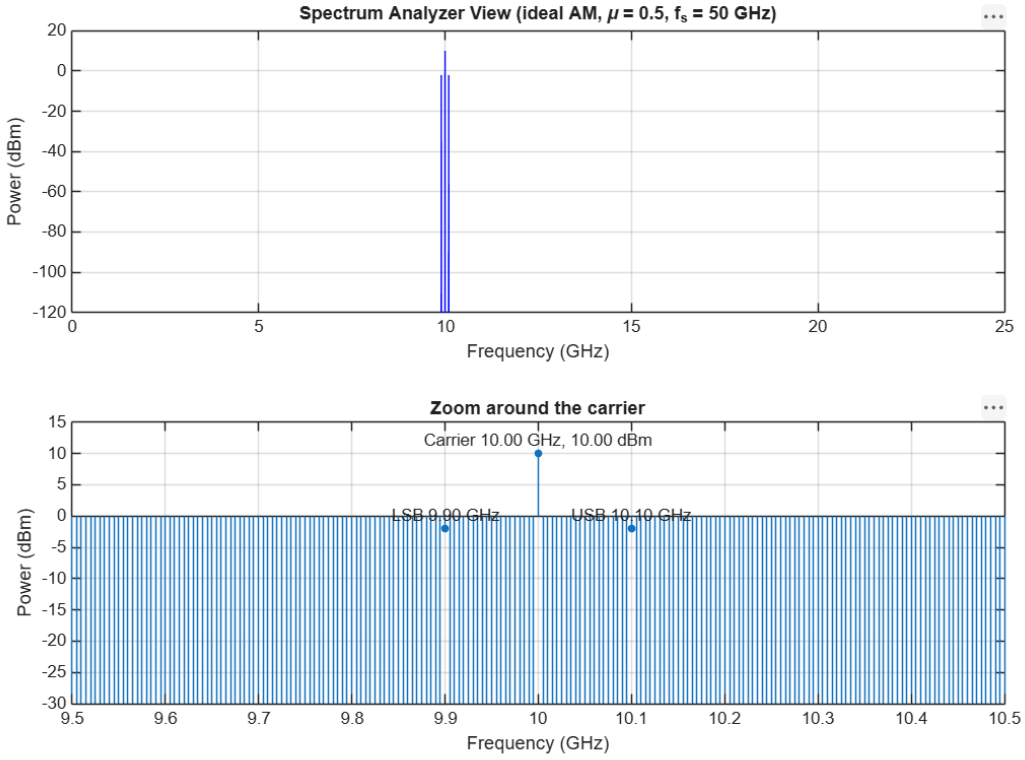
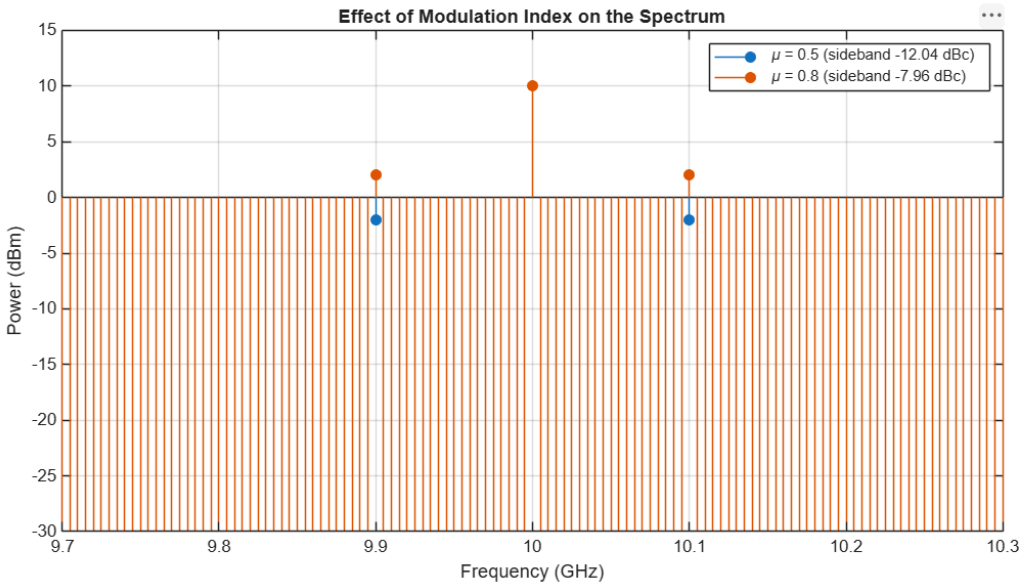
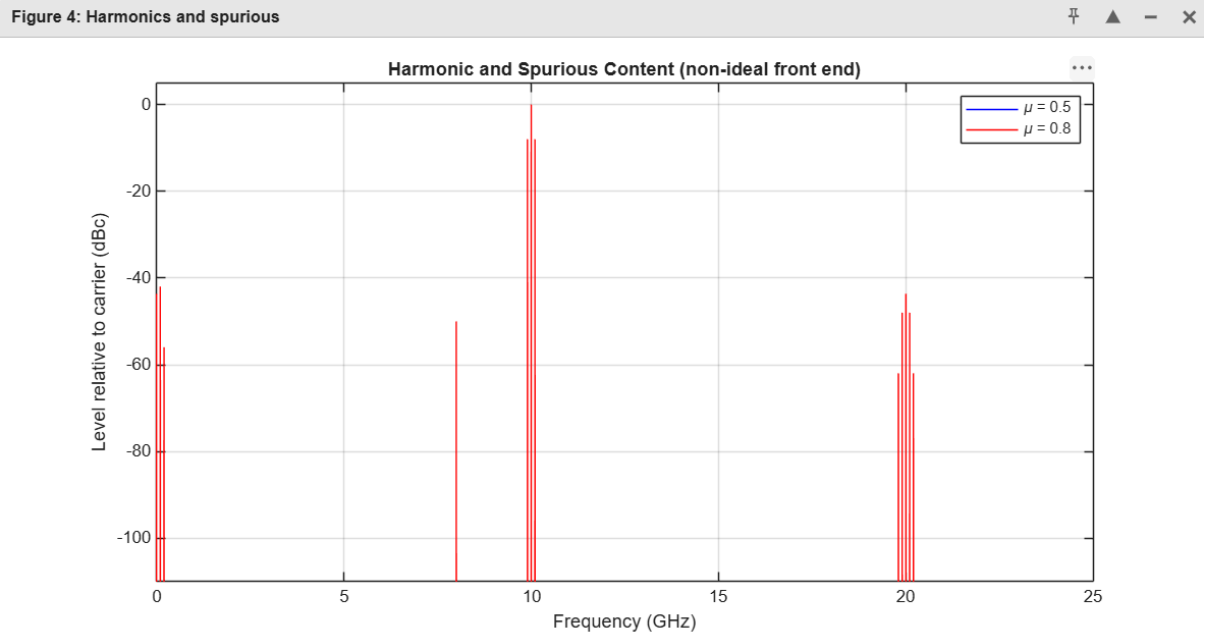
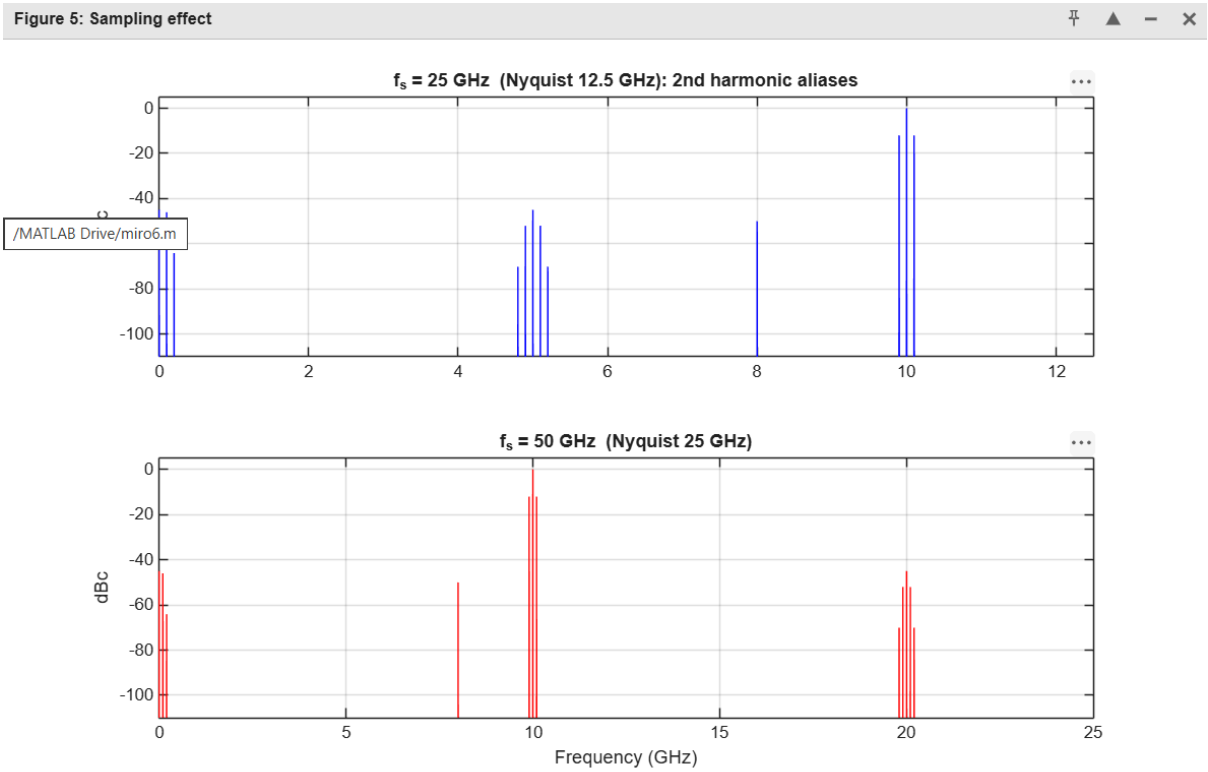


Figure 3: Modulation index effect







========== PART A : SPECTRUM ANALYSIS ==========
Samples N = 10000, resolution = 5.00 MHz, Nyquist = 25.0 GHz

Parameter Theory(GHz) Sim(GHz) %Dev
Carrier 10.0000 10.0000 0.0000
Lower sideband 9.9000 9.9000 0.0000
Upper sideband 10.1000 10.1000 0.0000

Carrier level : 10.00 dBm (1.0000 V peak)
Sideband level (each): -2.04 dBm (0.2500 V peak, theory mu*A/2 = 0.2500 V)
Sideband rel. carrier: -12.04 dBc (theory 20log10(mu/2) = -12.04 dBc)
Spurious level : < -150 dBc (numerical floor, ideal signal)

========== PART B : MODULATION-INDEX EFFECT ==========
Quantity mu = 0.5 mu = 0.8
Carrier amplitude (V) 1.0000 1.0000
Sideband amplitude (V) 0.2500 0.4000
Carrier level (dBm) 10.00 10.00
Sideband level (dBm) -2.04 2.04
Sideband level (dBc) -12.04 -7.96
Occupied BW = 2fm (GHz) 0.2000 0.2000
Total power (dBm) 10.51 11.21
Sideband power share (%) 11.11 24.24

Change in sideband level (mu 0.5 -> 0.8): +4.08 dB (theory 20log10(0.8/0.5) = 4.08 dB)
Carrier level change: 0.00 dB (carrier is independent of mu)

========== PART C : HARMONIC & SPURIOUS COMPONENTS ==========
Non-ideal model: a2 = 0.010, a3 = 0.000, spur tone 8.0 GHz at -50 dBc

--- mu = 0.5 ---
Carrier level : 10.00 dBm
Strongest spur: -45.00 dBc at 20.000 GHz -> Spur(dBc) = Pspur - Pc
n f_n = n*fc Apparent f Level(dBc)
2 20.00 20.00 -45.00
3 30.00 20.00 -45.00 (ALIASED)
Components above -70 dBc:
0.100 GHz -46.02 dBc
0.200 GHz -64.08 dBc
8.000 GHz -50.00 dBc
9.900 GHz -12.04 dBc
10.000 GHz 0.00 dBc
10.100 GHz -12.04 dBc
19.900 GHz -52.04 dBc
20.000 GHz -45.00 dBc
20.100 GHz -52.04 dBc

--- mu = 0.8 ---
Carrier level : 10.00 dBm
Strongest spur: -41.94 dBc at 0.100 GHz -> Spur(dBc) = Pspur - Pc

n f_n = n*fc Apparent f Level(dBc)
2 20.00 20.00 -43.61
3 30.00 20.00 -43.61 (ALIASED)
Components above -70 dBc:
0.100 GHz -41.94 dBc
0.200 GHz -55.92 dBc
8.000 GHz -50.00 dBc
9.900 GHz -7.96 dBc
10.000 GHz 0.00 dBc
10.100 GHz -7.96 dBc
19.800 GHz -61.94 dBc
19.900 GHz -47.96 dBc
20.000 GHz -43.61 dBc
20.100 GHz -47.96 dBc
20.200 GHz -61.94 dBc

Lower spur level (better spectral purity): mu = 0.5
Spec check (spur < carrier): PASS

========== PART D : SAMPLING-FREQUENCY EFFECT ==========
fs = 25 GHz : Nyquist 12.5 GHz, carrier 10.000 GHz, LSB 9.900, USB 10.100 GHz
fs = 50 GHz : Nyquist 25.0 GHz, carrier 10.000 GHz, LSB 9.900, USB 10.100 GHz
2nd harmonic (20 GHz): fs=25 GHz -> appears at 5.0 GHz (ALIASED); fs=50 GHz -> 20.0 GHz

========== OUTPUT PARAMETER TABLE ==========

Parameter	Theoretical	Simulated	PercentDeviation
{'Carrier frequency' }	10	10	0
{'Lower sideband' }	9.9	9.9	0
{'Upper sideband' }	10.1	10.1	0
{'Carrier level (dBm)' }	NaN	10	NaN
{'Spurious level (dBc)'}	NaN	-44.998	NaN

========== CONCLUSION ==========
The AM spectrum shows the carrier at 10.00 GHz and sidebands at 9.90 and 10.10 GHz, matching fc and fc +/- fm with 0.00 % deviation. Raising mu from 0.5 to 0.8 raises each sideband by 4.08 dB (-12.04 -> -7.96 dBc) while the carrier is unchanged and the occupied bandwidth stays 2fm = 0.2 GHz. Sidebands carry more power (11.1 % -> 24.2 %),

but non-ideal distortion also grows. Dominant unwanted component: -45.00 dBc at 20.00 GHz.

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